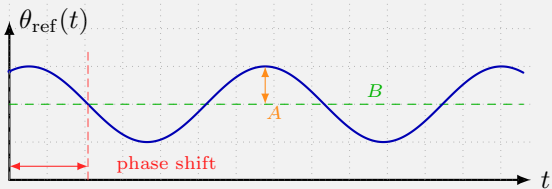


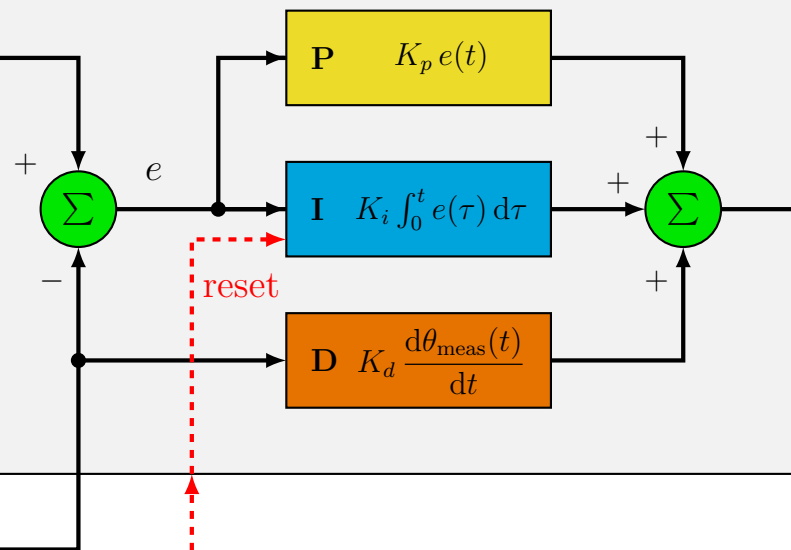
## Setpoint



$$\theta_{\text{ref}}(t) = B + A \sin(2\pi f t + \phi)$$

$\theta_{\text{ref}}$

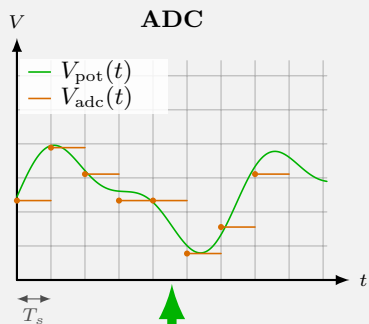
## PID Controller



contact event

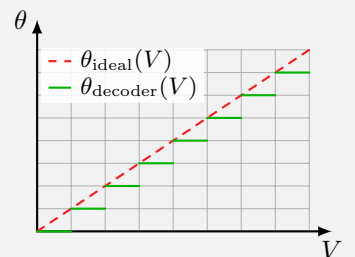
$u_{\text{control}}$

## Angle Sensor



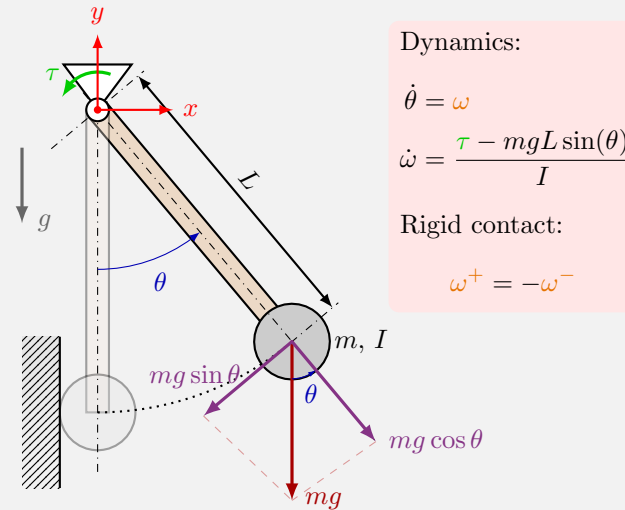
$V_{\text{adc}}$

## Decoder



$\theta_{\text{meas}}$

## Pendulum with Wall Contact



$\tau$

$\omega$

## BLDC Drive

